Bourgain-Rosenthal-Schechtman spaces in rearrangement-invariant and Hardy spaces

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Joint work with Konstantinos Konstantos and Pavlos Motakis

Brazilian Workshop on Banach Spaces

August 24, 2026

Introduction

- In 1981, Bourgain, Rosenthal, and Schechtman constructed uncountably many pairwise non-isomorphic *complemented* subspaces of L^p, $1 < p < \infty$, $p \neq 2$.
- Applications: e.g., number of closed operator ideals on L^p
- Ghawadrah 2016: extended results to reflexive Orlicz function spaces
- Konstantos-Motakis: factorization results in BRS spaces
- Goal: Extend original BRS result to more general rearrangement-invariant spaces and Hardy spaces.

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Definitions

- Dyadic intervals: $\mathcal{D} = \{[0,1), [0,\frac{1}{2}), [\frac{1}{2},1), [0,\frac{1}{4}), [\frac{1}{4},\frac{1}{2}), \dots\}$
- The Haar function h_I is 1 on the left and -1 on the right half of I.
- Together with $\mathbb{1}_{[0,1)}$, the Haar system $(h_I)_{I\in\mathcal{D}}$ is a monotone Schauder basis of L^p, $1 \le p < \infty$.
- Rademacher functions: $r_n = \sum_{|K|=2^{-n}} h_K$ for $n \in \mathbb{N}_0$.

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- A **Haar system Hardy space** Y is the completion of $\operatorname{span}\{h_I : I \in \mathcal{D}\}$ under a rearrangement-invariant norm $\|\cdot\|_X$ (i.e., $\operatorname{dist}(|f|) = \operatorname{dist}(|g|) \implies \|f\|_X = \|g\|_X$)

  or under the square function norm $\|\cdot\|_{H_X}$ given by

$$\Big\|\sum_{I\in\mathcal{D}} a_I h_I\Big\|_{H_X} = \Big\|\Big(\sum_{I\in\mathcal{D}} a_I^2 h_I^2\Big)^{1/2}\Big\|_X.$$

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Now fix a Haar system Hardy space Y.

- Construct R_α^Y, $0 \leq \alpha < \omega_1$ by alternately taking **disjoint** and **independent** sums:
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- Disjoint sum: $(Z \oplus W)_{\mathrm{disj}} = T_{[0,\frac{1}{2})}(Z) + T_{[\frac{1}{2},1)}(W)$, where $(T_{[a,b)} f)(t) = f(\frac{t-a}{b-a})$ on $[a,b)$
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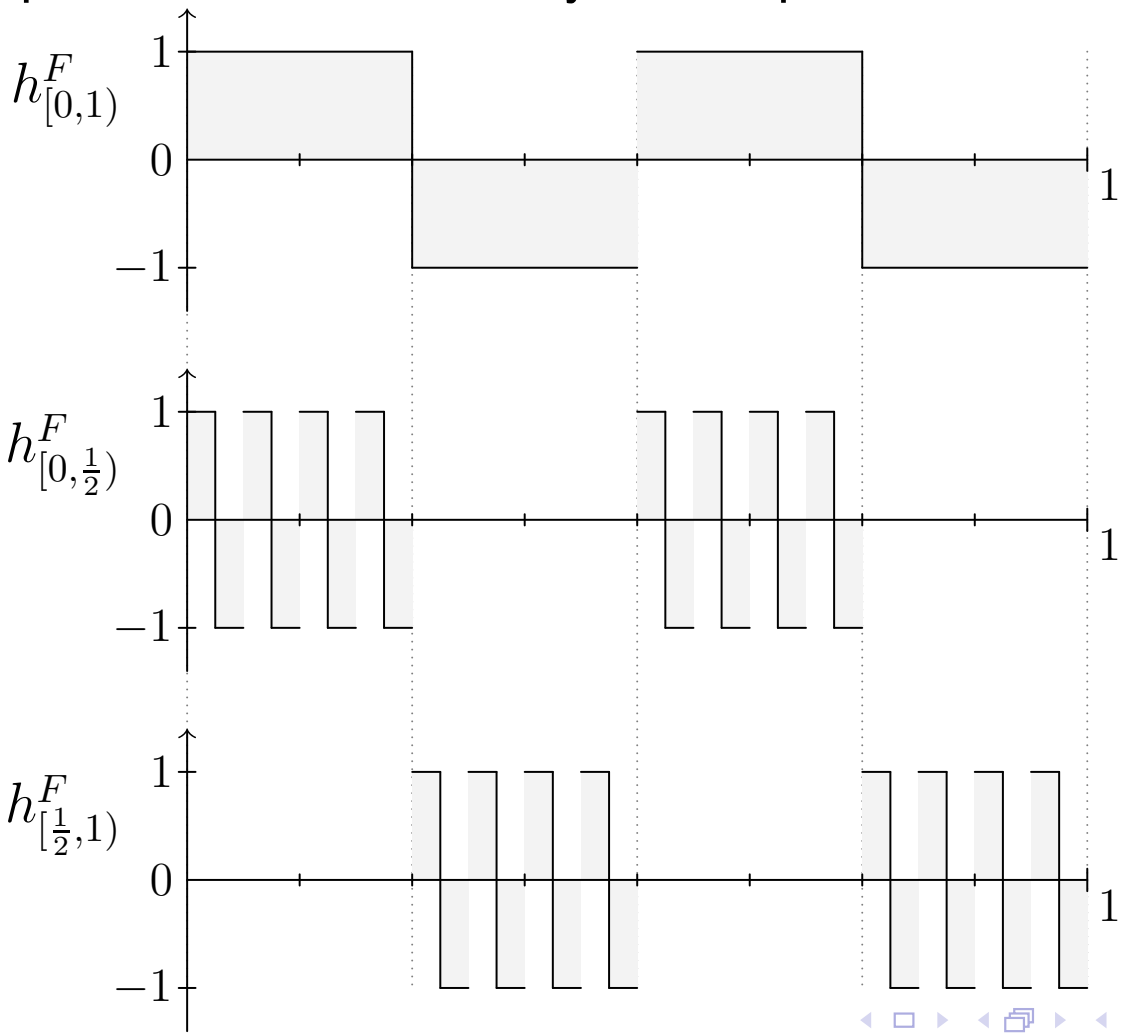
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- Now split $\mathbb{N}$ into infinitely many pairwise disjoint subsets $F_1, F_2, \ldots$.
- Map Z_n into Y_{F_n} via the above isometric embedding (call it J_n).
- Define $(\oplus_{n=1}^{\infty} Z_n)_{Y,\mathrm{ind}} = [J_n(Z_n)]_{n=1}^{\infty}$.

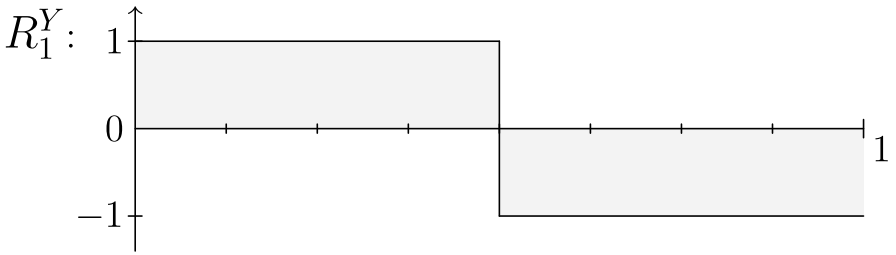
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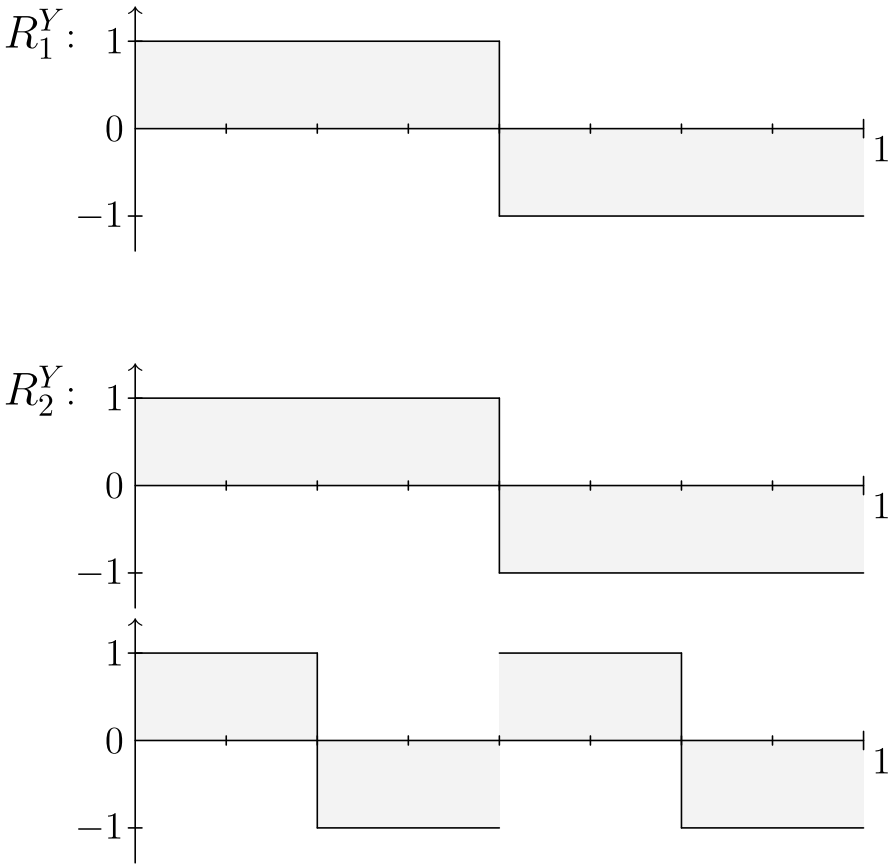
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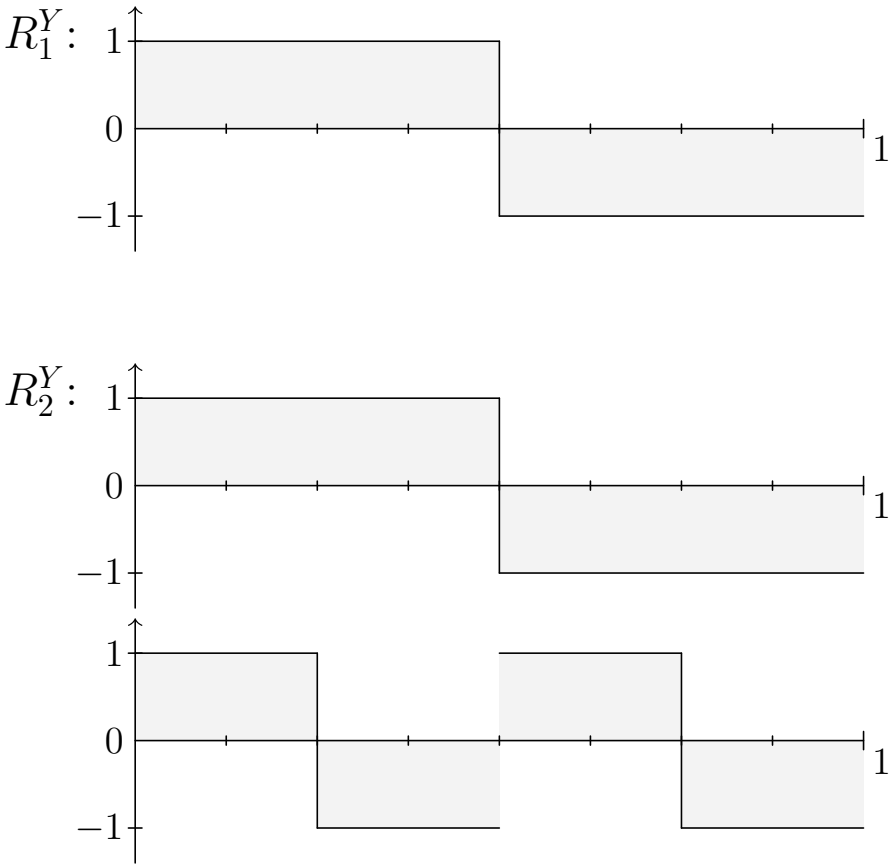
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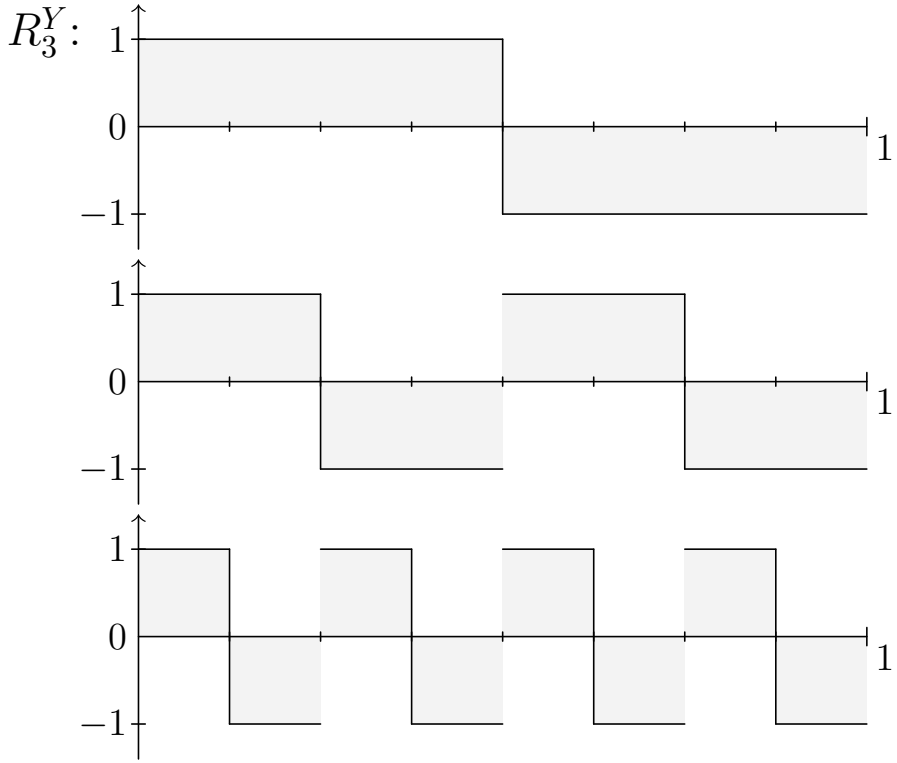


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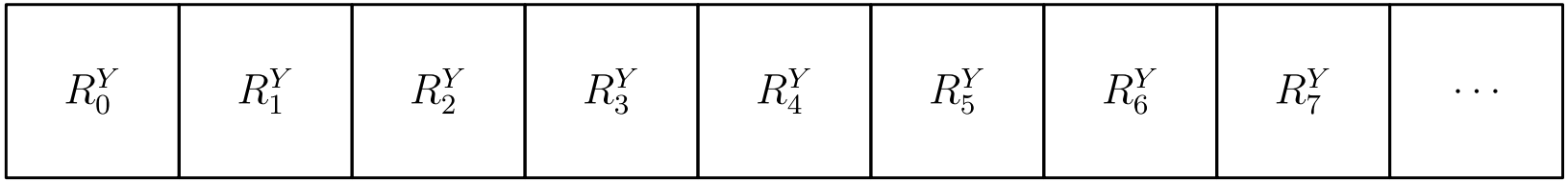


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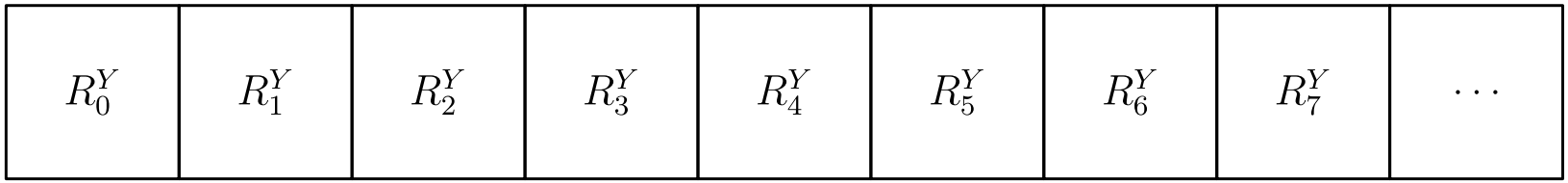




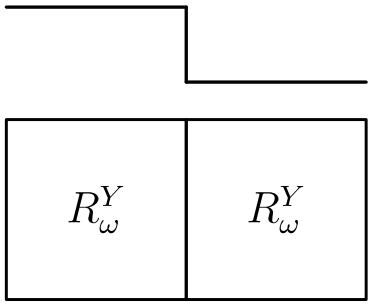
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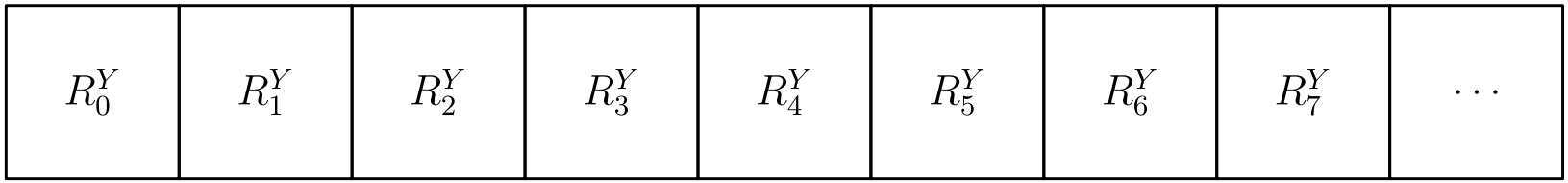
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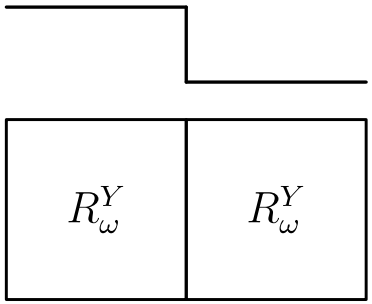
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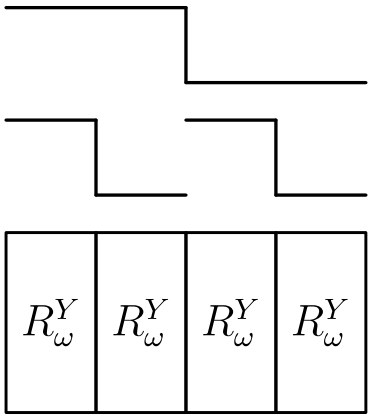
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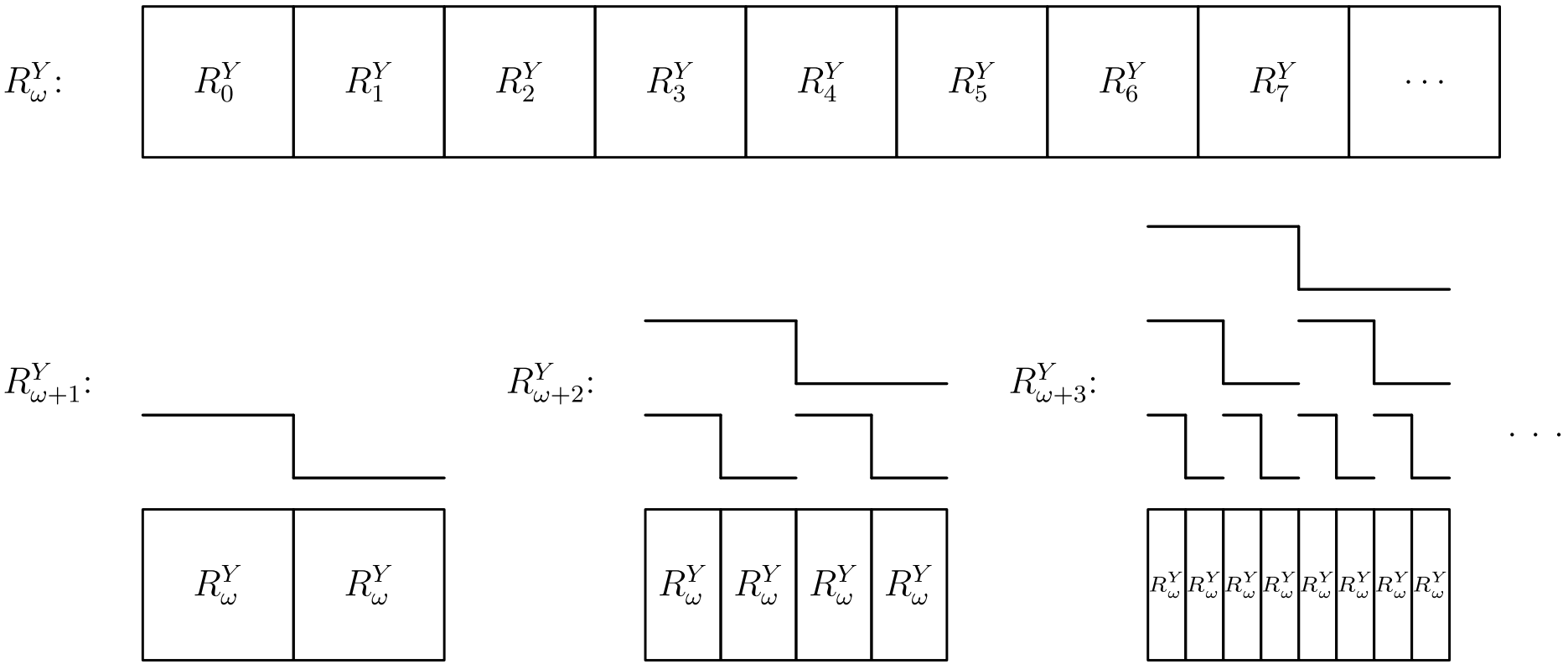


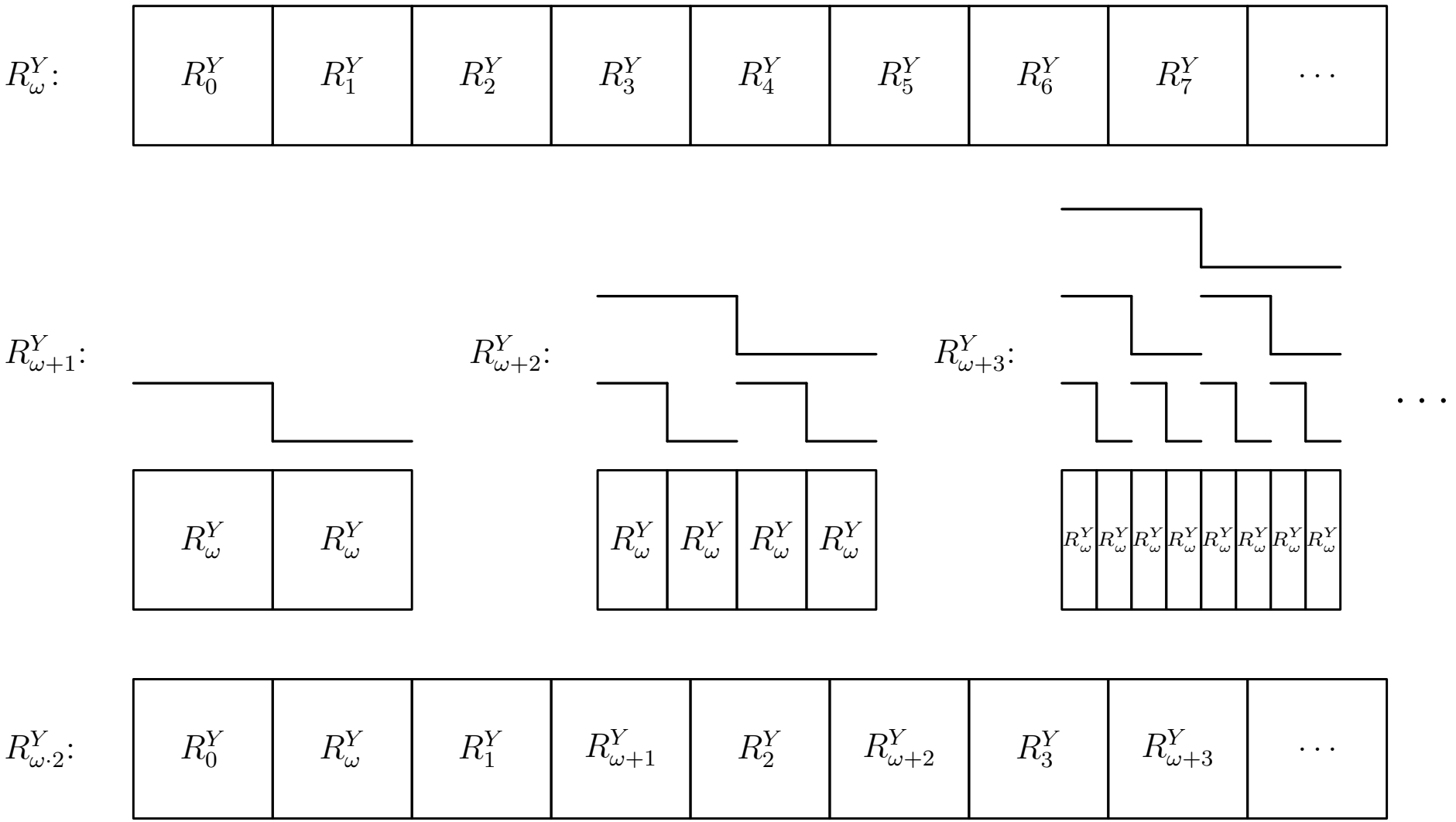
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Main result

Theorem (Konstantos, Motakis, S. '26)

Let X be a Haar system space with the Fatou property and non-trivial Boyd indices, and $Y = \{f \in X : \mathbb{E}f = 0\}$. Then all R_α^Y are complemented in Y, and one of the following exclusive alternatives holds:

- *There exists a strictly increasing transfinite sequence of countable ordinals $(\alpha_\xi)_{\xi<\omega_1}$ such that the spaces $R_{\alpha_\xi}^Y$, $\xi < \omega_1$, are pairwise non-isomorphic.*
- *For every ordinal $\omega \leq \alpha < \omega_1$, the space R_α^Y is isomorphic to Y.*

- Non-trivial Boyd indices = "not close to L^1 or L^∞".
- The second alternative is false in L^p, $1 < p < \infty$, except in L^2. Are there other examples where it is true?

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Theorem (Konstantos, Motakis, S. '26)

Let X be a Haar system space with the Fatou property and non-trivial Boyd indices, and $Y = \{f \in X : \mathbb{E}f = 0\}$. Then all R_α^Y are complemented in Y, and one of the following exclusive alternatives holds:

- *There exists a strictly increasing transfinite sequence of countable ordinals $(\alpha_\xi)_{\xi<\omega_1}$ such that the spaces $R_{\alpha_\xi}^Y$, $\xi < \omega_1$, are pairwise non-isomorphic.*
- *For every ordinal $\omega \leq \alpha < \omega_1$, the space R_α^Y is isomorphic to Y.*

- Non-trivial Boyd indices $=$ "not close to L^1 or L^∞".
- The second alternative is false in L^p, $1 < p < \infty$, except in L^2. Are there other examples where it is true?

Proof

- First, suppose that for some α, R_α^Y contains a complemented subspace isomorphic to Y.
- The *smallest* such α is a limit ordinal, so $Y \overset{c}{\hookrightarrow} (\oplus_{\beta<\alpha} R_\beta^Y)_{Y,\mathrm{ind}}$, but $Y \overset{c}{\not\hookrightarrow} R_\beta^Y$ for all $\beta < \alpha$.
- Ghawadrah 2016 $\implies$ there is an **independent** sequence in R_α^Y equivalent to the Haar system with closed span complemented in Y.
- Then such a system exists in any R_β^Y, $\omega \leq \beta < \omega_1$.
- Hence, $Y \overset{c}{\hookrightarrow} R_\beta^Y$, $R_\beta^Y \overset{c}{\hookrightarrow} Y$ (by Stein's inequality), and both are isomorphic to their squares.

  $\implies$ by Pełczyński's decomposition method, $Y \simeq R_\beta^Y$.

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Proof

- Now suppose that no space R_α^Y contains a complemented copy of Y.
- For every separable Banach space W, we define an ordinal index $i_Y[W] \in [0, \omega_1]$ such that:
  - $V \overset{c}{\hookrightarrow} W \implies i_Y[V] \leq i_Y[W]$
  - $Y \overset{c}{\hookrightarrow} W \iff i_Y[W] = \omega_1$.
- Thus, $i_Y[R_\alpha^Y] < \omega_1$ for all α.
- Now, for $\xi < \omega_1$, inductively choose α_ξ sufficiently large such that $i_Y[R_{\alpha_\xi}^Y] > \sup_{\eta<\xi} i_Y[R_{\alpha_\eta}^Y]$.
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Definition of complemented Y-index

For a separable Banach space W and $C \geq 1$, let $\mathscr{T}(Y, W, C)$ be the set of all finite sequences $\big((T_0, P_0), \ldots, (T_n, P_n)\big)$ satisfying:

- T_k is an embedding from $\operatorname{span}\{h_I\}_{|I|\geq 2^{-k}}$ into W; $\|T_k\|, \|T_k^{-1}\| \leq C$.
- P_k is a projection from W onto the image of T_k with norm $\leq C$.
- Some compatibility conditions.

$\mathscr{T}(Y, W, C)$ is a tree. The index $i_Y[W]$ is essentially the *order* of this tree as $C \to \infty$:

$$i_Y[W] := \begin{cases} \omega_1 & \text{if } \mathscr{T}(Y, W, C) \text{ ill-founded for some } C. \\ \sup_{C\geq 1} o[\mathscr{T}(Y, W, C)] & \text{otherwise.} \end{cases}$$

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Further results

Let $\Phi\colon [0,\infty) \to [0,\infty)$ be a strictly increasing Young's function with $\Phi(s) \sim s(\log s)^\alpha$ as $s \to \infty$, for some $0 < \alpha < 1/2$.

Theorem (Konstantos, Motakis, S. '26)

Let $Y = L^1_0$, H^1, the Orlicz space L^Φ_0, or H_{L^Φ}.

- *The space R^Y_α is uncomplemented in Y for $\omega \cdot 2 \leq \alpha < \omega_1$.*
- *For every $n \in \mathbb{N}_0$, there exist realizations of $R^Y_{\omega+n}$ that are uncomplemented in Y.*
- *But there exist realizations of $R^{H^1}_\omega$ that are complemented in H^1 (Müller-Schechtman 1989).*

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Thank you for your attention!